

FIFO AND LRU

```
Activities Terminal Mar 17 21:11
manishaa@lccalhost:~$ # FIFO (First In First Out)
# FIFO page replacement algorithm replaces the
# page that has been in memory the longest.
# It maintains a queue of pages based on arrival time.

1. When a page fault occurs and no free frame is available,
   remove the page that has been in memory the longest
   (the one at the front of the queue).
2. Add the new page to the end of the queue.
3. Simple to implement but may suffer from Belady's Anomaly.

# Example: Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3 -> Page Faults: 9

manishaa@lccalhost:~$
```

```
Activities Terminal Mar 17 21:11
manishaa@lccalhost:~$ # FIFO (First In First Out) - Example

# Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3

# Step-by-step (FIFO):
# Ref: 1 2 3 4 1 2 1 2 3 4 5
# Frames:
# [ 1 1 1 4 4 4 5 5 3 3 3 ]
# [ - 2 2 2 1 1 1 2 2 4 4 ]
# [ - - 3 3 3 3 2 2 2 2 5 ]

# Page Faults = 9

manishaa@lccalhost:~$
```

```
Activities Terminal Mar 17 21:11
manishaa@lccalhost:~$ # LRU (Least Recently Used)
# LRU page replacement algorithm replaces the page
# that has not been used for the longest time.
# It assumes that pages used in the recent past
# are likely to be used again in the near future.

1. When a page fault occurs and no free frame is available,
   remove the least recently used page.
2. Add the new page to memory and update its recent use.
3. More accurate than FIFO but more complex to implement.

# Example: Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3 -> Page Faults: 7

manishaa@lccalhost:~$
```

```
Activities Terminal Mar 17 21:11
manishaa@lccalhost:~$ # LRU (Least Recently Used) - Example

# Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
# Frames: 3

# Step-by-step (LRU):
# Ref: 1 2 3 4 1 5 1 2 3 4 5
# Frames:
# [ 1 1 1 4 4 5 5 5 3 3 3 ]
# [ - 2 2 2 2 2 1 1 1 4 4 ]
# [ - - 3 3 3 3 3 3 2 2 5 ]

# Page Faults = 7

manishaa@lccalhost:~$
```